Robust Email Spam Detection with Deep Learning:
Evaluating Classical and Neural Models on Public and Self-Created Emails

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Colab notebook: https://colab.research.google.com/drive/1QJJTQGDohRoLAtjUem0IS-3HbosLQ05s

1 Problem Description

Email remains one of the most heavily used communication channels in academic, professional, and personal life, which makes the detection of unwanted or malicious spam messages a critical task. Modern spam is highly diverse in both structure and linguistic expression, ranging from simplistic promotional content to carefully crafted phishing attempts that mimic legitimate correspondence. Effective spam filters must therefore go beyond simple keyword matching, and instead leverage lexical patterns, contextual cues, and even higher-level document structure.

In this project we compare three modeling paradigms for email spam detection: a sparse bag-of-words / TF–IDF representation with Logistic Regression, a fine-tuned BERT transformer, and a Hierarchical Attention Network (HAN) built on word2vec embeddings. Our primary goal is both to achieve high in-distribution performance, and to study how these models behave under distribution shift. To this end, we first train and validate on the Kaggle Spam Email Dataset, and then evaluate all models on a self-created Gmail-style test set collected by the authors. This custom test set differs substantially in writing style, topic, and formatting from the Kaggle data, and thus serves as a more realistic proxy for deployment conditions. By comparing classic linear models to large pre-trained transformers and hierarchical neural architectures on both datasets, we aim to understand the trade-offs between accuracy, robustness, and model complexity in real-world spam filtering.

2 Approach

2.1 Data Processing

We use two sources of email data. The first is the Kaggle Spam Email Dataset, which contains labeled spam and ham messages in a standardized format. We apply an 80/20 stratified split to create training and validation sets. **Our test dataset is a self-created Gmail corpus containing 843 real messages, relatively balanced between spam and ham, collected from project members' real email accounts.** These emails vary in style, structure, and content, and therefore provide a more realistic and challenging evaluation setting.

To address class imbalance in the Kaggle training split, we apply random upsampling of the minority class, while leaving all test sets unchanged. Concretely, spam examples are sampled with replacement until the spam and ham classes are approximately balanced in the training data. This helps the models avoid a trivial bias toward the majority (ham) class and encourages them to learn features that are discriminative for spam. We do not modify the distribution of the validation or self-created Gmail test sets, so that all reported metrics reflect performance under the original proportions.

2.2 Models

Bag-of-Words / TF–IDF + Logistic Regression

As a classical baseline, we use sparse bag-of-words representations with Logistic Regression. Concretely, we first construct a standard Bag-of-Words (BoW) model based on token counts, and then apply TF–IDF reweighting on top of the same vocabulary. Both BoW and TF–IDF features are fed into a Logistic Regression classifier with class weighting to compensate for label imbalance. These linear models rely purely on word-frequency statistics and ignore word order, but they are simple, fast to train, and often strong baselines when lexical distinctions between spam and ham are pronounced.

In addition, we construct sparse Bag-of-Words representations using unigram, bigram, and trigram features, so the model can capture not only individual tokens but also short phrases characteristic of spam (e.g., "free trial", "limited time offer", "click here now"). Concretely, we build a vocabulary over all 1–3 gram patterns in the training data and represent each email as a high-dimensional count vector. The TF–IDF features used in our main baseline are obtained by reweighting these same n-gram counts by term frequency and inverse document frequency, so TF–IDF can be viewed as a rescaled Bag-of-Words model rather than a fundamentally different representation.

BERT

As a deep contextual baseline, we fine-tune the `bert-base-uncased` model. Each email is tokenized into subwords, truncated or padded to a maximum length of 256, and fed through the transformer encoder. A classification layer attached to the `[CLS]` token is trained using AdamW. Unlike the BoW/TF–IDF models, which treat emails as unordered bags of words, BERT models word order and context, making it well suited for capturing subtler spam patterns such as phrasing, style, and cross-sentence cues.

Hierarchical Attention Network

Our third approach is a Hierarchical Attention Network (HAN), which explicitly models document structure. We split each email into subject and body, apply sentence tokenization to the body, and clean and tokenize each sentence into words. A 200-dimensional Word2Vec model trained on the corpus provides word embeddings, so the representation is no longer sparse BoW counts but dense, learned vectors. Each email becomes a padded tensor of shape `(sentences` $\times$ `words)`. A HAN model encodes word sequences using a BiGRU with word-level attention, then encodes sentence vectors with another BiGRU and sentence-level attention. This architecture captures the hierarchical organization of emails (words $\rightarrow$ sentences $\rightarrow$ document) and, unlike BoW/TF–IDF, offers interpretability via attention weights over both words and sentences.

3 Results

Model	Acc.	Ham P	Ham R	Ham F1	Spam P	Spam R	Spam F1
TF–IDF (1–2-gram) + LR	0.963	0.98	0.95	0.96	0.95	0.98	0.96
BoW (1-gram) + LR	0.956	0.94	0.97	0.96	0.97	0.94	0.96
BoW (1–2-gram) + LR	0.951	0.93	0.97	0.95	0.97	0.93	0.95
BoW (1–3-gram) + LR	0.953	0.93	0.97	0.95	0.97	0.93	0.95
BERT	0.910	0.87	0.96	0.91	0.95	0.86	0.91
HAN (best threshold)	0.980	0.99	0.99	0.99	0.97	0.96	0.97

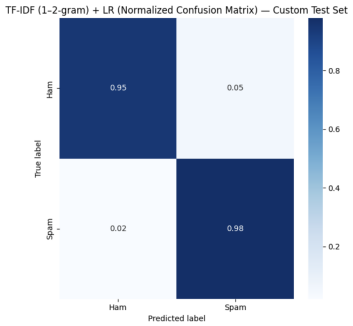
Table 1: Performance on the self-created email test set (P = precision, R = recall)

TF–IDF + Logistic Regression

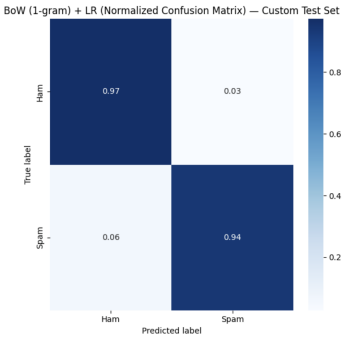
The result shows the normalized confusion matrix for the TF–IDF baseline. On the Kaggle dataset the model achieves roughly 99% accuracy, with F1-scores of 0.99 for ham and 0.98 for spam, indicating that a simple linear classifier over n-gram TF–IDF features is already sufficient to separate most messages. On the custom Gmail test set the performance drops slightly (Table 1), but TF–IDF + LR still attains mid-90s F1 on both classes. Most of the remaining errors are spam messages that are labeled as ham, suggesting that some phishing or promotional emails in the custom set use vocabulary that is closer to genuine correspondence than to the Kaggle spam examples.

Bag-of-Words (1–3-gram)

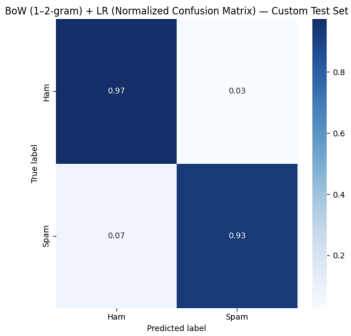
For comparison, we also evaluate Bag-of-Words models using 1–3-gram counts with Logistic Regression (Figure 1). On both datasets these models achieve accuracies in the mid-to-high 90% range, with spam F1-scores slightly below those of the TF–IDF baseline. Using higher-order n-grams marginally improves precision on spam but does not close the gap to TF–IDF, and sometimes hurts recall. Overall, the BoW models confirm that simple lexical statistics already capture a large fraction of the signal, but TF–IDF weighting provides a more robust representation than raw count features.



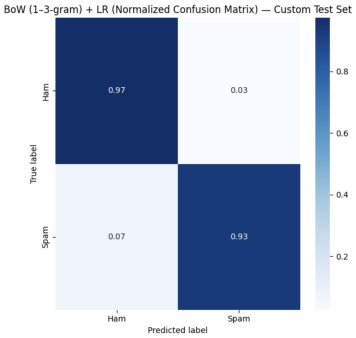
(a) TF–IDF (1–2-gram) + LR.



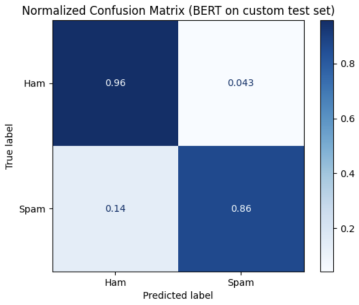
(b) BoW (1-gram) + LR.



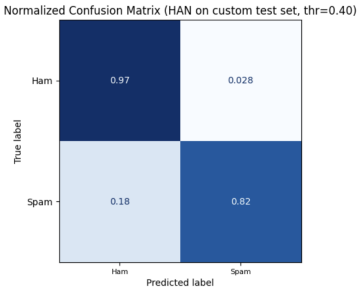
(c) BoW (1–2-gram) + LR.



(d) BoW (1–3-gram) + LR.



(e) BERT classifier.



(f) Hierarchical Attention Network (HAN).

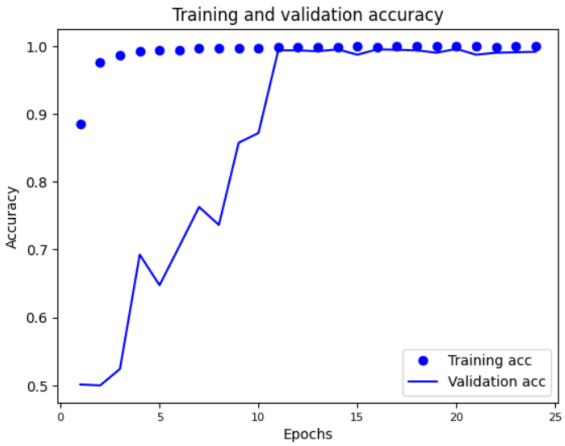
Figure 1: Normalized confusion matrices for all models on the self-created email test set

BERT

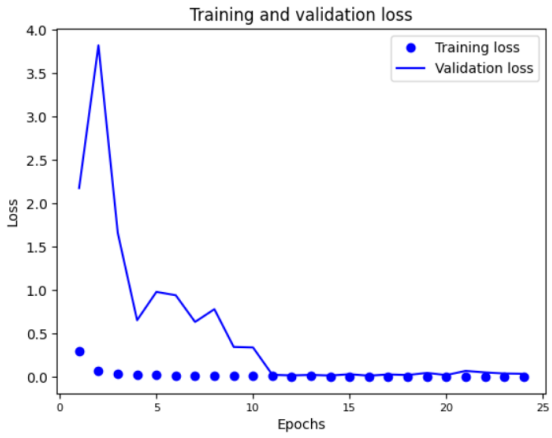
BERT demonstrates strong performance on the Kaggle data: the confusion matrices in Figure 1e show an accuracy of about 0.99 with balanced errors across ham and spam. However, on the custom Gmail set accuracy drops to around 0.91, and the per-class metrics reveal a clear asymmetry: ham recall remains high, while spam recall is noticeably lower. In other words, BERT tends to miss more spam messages when evaluated out of domain. This suggests that, despite its contextual modeling capacity, the fine-tuned BERT model has partially overfit to the style and distribution of the Kaggle corpus and does not fully transfer to our self-created Gmail data.

Hierarchical Attention Network

Training curves for the HAN model are displayed in Figure 2, and confusion matrices for both datasets appear in Figure 1f. On Kaggle, HAN attains an accuracy of about 0.98, comparable to BERT and only slightly below the TF–IDF baseline. On the custom Gmail dataset, performance decreases to roughly 0.91, with the primary degradation occurring in spam recall. The confusion matrix shows that HAN remains very conservative on ham but fails to flag a larger fraction of spam emails. This pattern is consistent with a model that has learned to exploit structural regularities in the Kaggle messages (e.g., subject–body layout and sentence structure) that do not generalize perfectly to the more diverse Gmail corpus.



(a) Training accuracy



(b) Training loss

Figure 2: Training accuracy and loss of the HAN model

4 Lessons Learned

Performance Under Distribution Shift

Our study highlights an important distinction between achieving high accuracy on benchmark datasets and building models that perform reliably in real-world email environments. All three approaches (TF–IDF + Logistic Regression, BERT, and HAN) achieve near-perfect performance on the Kaggle dataset, but their behavior diverges substantially when evaluated on our self-created Gmail-style corpus. This distribution shift mirrors the variability that email providers face in production systems, where message length, formatting, tone, and writing style can differ widely across users and contexts.

Why Simple Models Generalize Better

From a practical standpoint, the most robust model was also the simplest. TF–IDF + Logistic Regression, despite its lack of contextual or hierarchical representation, preserved mid-90s F1 scores under distribution shift. This suggests that many spam indicators that persist across domains are encoded in stable lexical patterns—specific tokens, short phrases, and frequency-based cues that generalize well from Kaggle to real inbox traffic. In contrast, BERT and HAN, both of which introduce far greater modeling capacity, experienced sharper drops in spam recall on the Gmail set. This degradation in out-of-domain performance reflects a common challenge in production ML systems: high-capacity models can overfit benchmark-specific stylistic regularities and fail to adapt when confronted with naturally occurring variation.

Challenges in Real-World Deployment

Several factors help explain this performance degradation under domain shift. The Kaggle data is relatively small for fine-tuning a 110M-parameter transformer or training a hierarchical network from scratch, making these models prone to overfitting. Preprocessing steps also created structural mismatches: BERT truncates emails to 256 tokens, while HAN depends on clean sentence boundaries and embeddings learned exclusively from Kaggle. In contrast, real-world emails contain signatures, embedded replies, URLs, inconsistent punctuation, and diverse formatting—all of which violate assumptions built into deeper models. These challenges highlight why many industrial spam filters rely on hybrid or lexical-first approaches rather than purely end-to-end neural architectures. More broadly, data scarcity and domain shift remain persistent obstacles. High-quality labeled email corpora are difficult to obtain due to privacy constraints, and spam content evolves continuously, creating a moving target for any classifier. As a result, robustness, maintainability, and ease of retraining can be as important as raw accuracy.

5 Conclusion

This project systematically evaluated classical lexical models and modern neural architectures for email spam detection under both benchmark and real-world conditions. All models achieved near-perfect performance on the Kaggle dataset, but their behavior diverged substantially under distribution shift to a self-created Gmail-style corpus. TF–IDF with Logistic Regression emerged as the most robust approach, maintaining strong and balanced performance across domains, whereas BERT and the Hierarchical Attention Network suffered notable drops in spam recall despite their greater representational capacity. These results underscore that model complexity alone does not guarantee generalization and that stable lexical cues remain highly effective for spam detection in practice. Overall, our findings highlight the importance of out-of-distribution evaluation and demonstrate that simplicity, robustness, and ease of adaptation are critical considerations when deploying spam filters in real-world email systems.

Opportunities for Future Improvement

Our findings point to several directions for strengthening robustness and generalization. Smaller pretrained models such as DistilBERT or domain-adapted transformer variants could reduce overfitting in low-data settings. More extensive data cleaning and normalization, especially around signatures, URLs, and HTML artifacts, may help bridge stylistic differences between email sources. Incorporating additional signals such as sender metadata, URL reputation, or ensemble combinations of lexical and contextual models would further align our system with how commercial filters operate. Finally, expanding our dataset, particularly with diverse unlabeled emails, would enable domain-adaptive pretraining strategies known to improve NLP model transferability.

Beyond modeling and data considerations, future work could also investigate evaluation and deployment-oriented aspects that are not captured by standard classification metrics. Analyzing error costs in an asymmetric manner, in which false negatives (missed spam or phishing) may be far more harmful than false positives, would better reflect real-world priorities. Incorporating human-in-the-loop feedback mechanisms, such as user reports or active learning strategies, could further improve adaptability over time.